Base station physics: summary

Params: base/spec/params.py v0.1-stub (still defaults for LAUNCH, AIM, REEL, LAYOUT)

Spring (each, A228, SF 1.2 on Shigley static limits)

Rate used / by formula	0.125 / 0.128 N/mm
Initial tension used	7.0 N
Body coils, mass	98, 55 g
Safe force (governed by)	43.6 N (hook torsion)
Safe extension (wire) / spec cap	293 / 250 mm
Surge (fixed-fixed)	24 Hz

Launch (chain spring model)

Stroke, ratio, pitch	240 mm, 1.0:1, 35 deg
Spring extension full / rest	250 / 10 mm
Stored energy (2 springs)	11.2 J
Carriage force at full draw	76 N
Muzzle speed	7.13 m/s
Peak acceleration, stroke time	20 g, 56 ms
Efficiency (dart KE / stored)	63 %
Carriage KE at stop, pad peak	1.52 J, 762 N (4 mm pad)
Spring KE at stop, coil clash	0.99 J, 10.5 m/s

Flight (bead-chain tether, level ground)

Range at 0.5 / 0.75 / 1.0 N brake	4.20 / 3.77 / 3.40 m (pitch 35)
Same shot, no tether	5.22 m
Flight time, apex (0.75 N)	0.81 s, 1.07 m
Impact speed, angle (0.75 N)	5.8 m/s, 54 deg down
Ballistics table	15 to 55 deg, best 35.0 deg: 3.96 m

Storage reel payout (0.75 N at full reel)

Reel empty mass, inertia full	35 g, 19.4 kg mm2
Line radius full / core	27.4 / 20.0 mm (brake +37 % at core)
Line pull in the stroke (spin-up)	4.00 N
Reel rpm max	2483 rpm
Overrun after landing	0.23 m in 0.12 s

Capstan and tow

Wraps for tow limit (mu 0.15)	3.4 raw, 6 design
Take-up needed (design wraps)	0.45 N
Tow at limit: torque, speed, current	1.19 N m, 0.077 m/s, 2.80 A
Line pull at wiper stall (12.6 V)	597 N (break 667 N)

Drivetrain (wiper at 11.1 V with sag)

aim: peak torque, current, time	0.19 N m, 1.7 A, 5 s
cock: peak torque, current, time	1.29 N m, 2.9 A, 4 s
reel: peak torque, current, time	0.08 N m, 1.6 A, 235 s
tow: peak torque, current, time	1.19 N m, 2.8 A, 65 s
Wiper stall 12.6 V (with sag)	14.1 N m, 17.0 A

Harry must measure (then edit the parameter block and rerun: python summary.py)

- Spring hang test: hang 1, 2, 3 kg from one spring, measure the length each time. Slope = rate, intercept = initial tension. Every range number scales with the stored energy.
- Wiper: free speed (turns in 30 s at 12.0 V), free current, stall torque (arm + luggage scale, 1 s) and stall current. Edit common.Wiper or spec WIPER.
- Dart mass on a kitchen scale with ballast and tether knot (spec 0.279 kg).
- Slip clutch: pull the line off the reel with a luggage scale; set 0.5 to 0.75 N.
- Rail friction: tilt the rail with the dart on it until it slides (mu = tan angle).